

基于 Hu-Washizu 变分原理的薄壳结构 准变分一致无网格法

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薄壳结构控制方程

壳体中面应变与曲率

$$\begin{aligned}\epsilon_{\alpha\beta} &= \frac{1}{2}(\mathbf{a}_\alpha \cdot \mathbf{u}_{,\beta} + \mathbf{u}_{,\alpha} \cdot \mathbf{a}_\beta) \\ &= \epsilon_{\alpha\beta} + \kappa_{\alpha\beta} \xi^3\end{aligned}$$

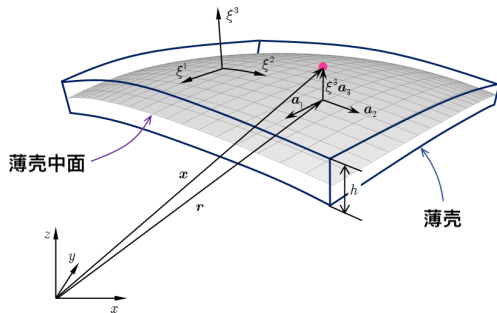
Kirchhoff 假设

$$\epsilon_{3i} = 0 \Rightarrow \Rightarrow \boldsymbol{\theta} = -\mathbf{u}_{,\alpha} \cdot \mathbf{a}_3 \mathbf{a}^\alpha$$



薄壳结构曲率

$$\begin{aligned}\epsilon_{\alpha\beta} &= \frac{1}{2}(\mathbf{a}_\alpha \cdot \mathbf{u}_{,\beta} + \mathbf{u}_{,\alpha} \cdot \mathbf{a}_\beta) \\ \kappa_{\alpha\beta} &= (\Gamma_{\alpha\beta}^\gamma \mathbf{u}_{,\gamma} - \mathbf{u}_{,\alpha\beta}) \cdot \mathbf{a}_3 = -\mathbf{u}_{,\alpha|\beta} \cdot \mathbf{a}_3\end{aligned}$$



薄壳结构曲面

薄壳结构分析曲率近似需要高阶形函数

再生核无网格近似

位移近似

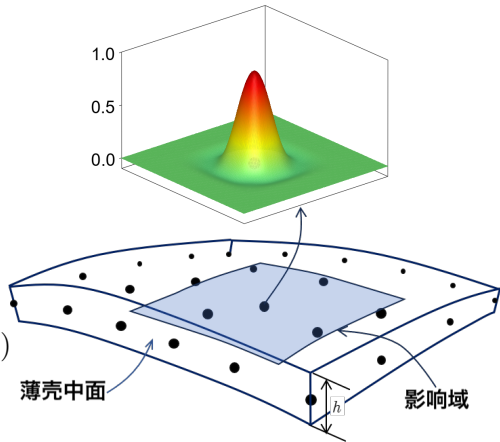
$$u^h(\xi) = \sum_{I=1}^{n_p} \Psi_I(\xi) d_I$$

$$\Psi_I(\xi) = p^T(\xi) c(\xi) \phi(\xi_I - \xi)$$

一致性条件

$$\sum_{I=1}^{n_p} \Psi_I(\xi) p(\xi_I) = p(\xi) \Rightarrow c(\xi) = A^{-1}(\xi) p(0)$$

$$\Psi_I(\xi) = p^T(\xi_I - \xi) A^{-1}(\xi) p(0) \phi(\xi_I - \xi)$$

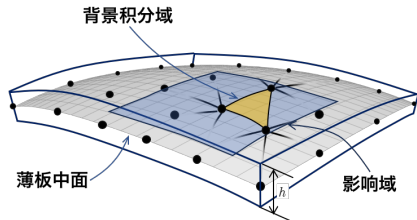


再生核近似

变分一致性

变分一致性阶次等于伽辽金弱形式能准确求解由多项式所组成的精确解阶次:

$$\int_{\Omega} h \delta \varepsilon_{\alpha\beta} C^{\alpha\beta\gamma\eta} \varepsilon_{\gamma\eta} d\Omega + \int_{\Omega} \frac{h^3}{12} \delta \kappa_{\alpha\beta} C^{\alpha\beta\gamma\eta} \kappa_{\gamma\eta} d\Omega = f(p^{[q]})$$



伽辽金无网格法数值积分

伽辽金无网格法误差估计 (Wu and Wang, 2021)

$$\|u - u_h\|_{H_e} \leq \underbrace{C_p s^p |u|_{H_p}}_{\text{一致性条件控制}} + \underbrace{C_q s^q |u|_{H_q}}_{\text{变分一致性条件控制}}$$

变分一致型边界条件施加方案

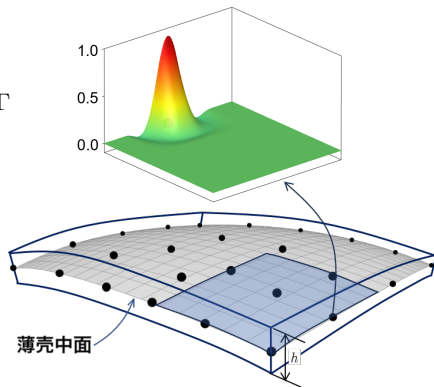
Nitsche 法施加边界条件

- 稳定项

$$K_{IJ}^s = \alpha_v \int_{\Gamma_v} \Psi_I \Psi_J d\Gamma + \alpha_\theta \int_{\Gamma_\theta} \Psi_{I,\eta} n^\eta \mathbf{a}_3 \mathbf{a}_3 n^\gamma \Psi_{J,\gamma} d\Gamma \\ + \alpha_C [[\Psi_I \mathbf{a}_3 \mathbf{a}_3 \Psi_J]]_{\mathbf{x} \in C_v}$$

- 一致项

$$K_{IJ}^c = - \int_{\Gamma_v} (\Psi_I \mathbf{T}_{NJ}(\Psi_{J,\alpha\beta\gamma}) + \mathbf{T}_{NI}(\Psi_{I,\alpha\beta\gamma}) \Psi_J) d\Gamma \\ + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma \mathbf{a}_3 M_{nnJ} + \mathbf{a}_3 M_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ + ([[\Psi_I \mathbf{a}_3 \mathbf{P}_J]] + [[\mathbf{P}_I \mathbf{a}_3 \Psi_J]])_{\mathbf{x} \in C_v}$$



边界点处无网格形函数

变分一致型边界条件施加方案

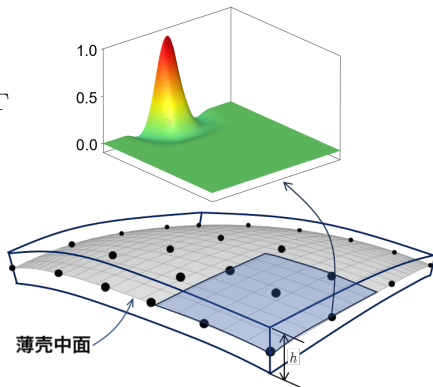
Nitsche 法施加边界条件

- 稳定项 $\times$ 人工经验参数影响求解精度

$$K_{IJ}^s = \alpha_v \int_{\Gamma_v} \Psi_I \Psi_J d\Gamma + \alpha_\theta \int_{\Gamma_\theta} \Psi_{I,\eta} n^\eta a_3 a_3 n^\gamma \Psi_{J,\gamma} d\Gamma \\ + \alpha_C [[\Psi_I a_3 a_3 \Psi_J]]_{x \in C_v}$$

- 一致项

$$K_{IJ}^c = - \int_{\Gamma_v} (\Psi_I \mathbf{T}_{NJ}(\Psi_{J,\alpha\beta\gamma}) + \mathbf{T}_{NI}(\Psi_{I,\alpha\beta\gamma}) \Psi_J) d\Gamma \\ + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma a_3 M_{nnJ} + a_3 M_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ + ([[\Psi_I a_3 \mathbf{P}_J]] + [[\mathbf{P}_I a_3 \Psi_J]])_{x \in C_v}$$



边界点处无网格形函数

变分一致型边界条件施加方案

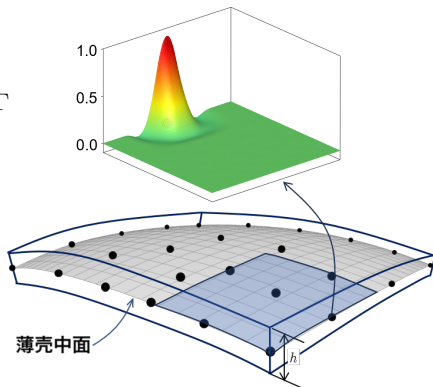
Nitsche 法施加边界条件

- 稳定项 × 人工经验参数影响求解精度

$$K_{IJ}^s = \alpha_v \int_{\Gamma_v} \Psi_I \Psi_J d\Gamma + \alpha_\theta \int_{\Gamma_\theta} \Psi_{I,\eta} n^\eta a_3 a_3 n^\gamma \Psi_{J,\gamma} d\Gamma \\ + \alpha_C [[\Psi_I a_3 a_3 \Psi_J]]_{x \in C_v}$$

- 一致项 × 需计算形函数高阶导数

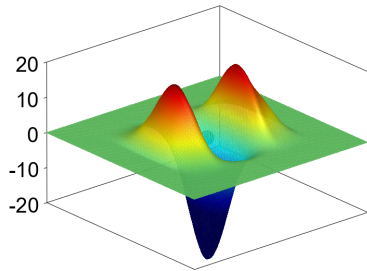
$$K_{IJ}^c = - \int_{\Gamma_v} (\Psi_I \mathbf{T}_{NJ}(\Psi_{J,\alpha\beta\gamma}) + \mathbf{T}_{NI}(\Psi_{I,\alpha\beta\gamma}) \Psi_J) d\Gamma \\ + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma a_3 M_{nnJ} + a_3 M_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ + ([[\Psi_I a_3 \mathbf{P}_J]] + [[\mathbf{P}_I a_3 \Psi_J]])_{x \in C_v}$$



边界点处无网格形函数

无网格形函数高阶导数

$$\Psi_{I,ij}(\xi) = \begin{pmatrix} p_{,ij}^T(\xi_I - \xi)A^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p_{,i}^T(\xi_I - \xi)A_{,j}^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p_{,i}^T(\xi_I - \xi)A^{-1}(\xi)\phi_{s,j}(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A_{,ij}^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p_{,j}^T(\xi_I - \xi)A_{,i}^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A_{,i}^{-1}(\xi)\phi_{s,j}(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A^{-1}(\xi)\phi_{s,ij}(\xi_I - \xi) \\ +p_{,j}^T(\xi_I - \xi)A^{-1}(\xi)\phi_{s,i}(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A_{,j}^{-1}(\xi)\phi_{s,i}(\xi_I - \xi) \end{pmatrix} p(0)$$



形函数二阶导数

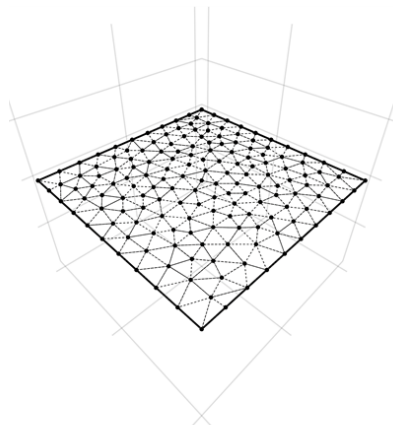
无网格形函数高阶导数
计算复杂耗时

基于 Hellinger-Reissner 变分一致型伽辽金无网格法

薄壳伽辽金法的变分一致性

$$\begin{array}{ccc} u(\xi) = p^T(\xi)a & \Rightarrow & \begin{array}{l} \varepsilon_{\alpha\beta} = \frac{1}{2}(\mathbf{a}_\alpha \cdot \mathbf{u}_{,\beta} + \mathbf{u}_{,\alpha} \cdot \mathbf{a}_\beta) \\ \kappa_{\alpha\beta} = -\mathbf{u}_{,\alpha}|_\beta \cdot \mathbf{a}_3 \end{array} \\ \text{多项式} & & \text{非多项式} \end{array}$$

数值算例



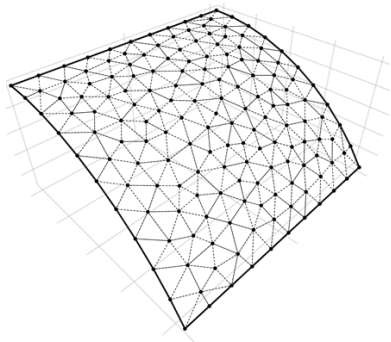
Flat model

Results of patch test for flat model.

	Linear patch test		Quadratic patch test	
	L_2 -Error	H_e -Error	L_2 -Error	H_e -Error
GI-Penalty	1.41E-04	4.62E-03	1.97E-03	7.17E-03
GI-Nitsche	1.73E-04	5.61E-03	1.85E-03	7.76E-03
RKGSi-Penalty	5.04E-09	1.02E-07	3.01E-09	3.41E-08
RKGSi-Nitsche	9.75E-12	8.98E-11	1.29E-12	1.06E-12
RKGSi-HR	6.15E-13	6.91E-12	7.51E-13	8.36E-12

$$v = \begin{Bmatrix} (\xi^1 + 2\xi^2)^n \\ (3\xi^1 + 4\xi^2)^n \\ (5\xi^1 + 6\xi^2)^n \end{Bmatrix}, \quad n = \begin{cases} 1 & \text{Linear} \\ 2 & \text{Quadratic} \end{cases}$$

数值算例



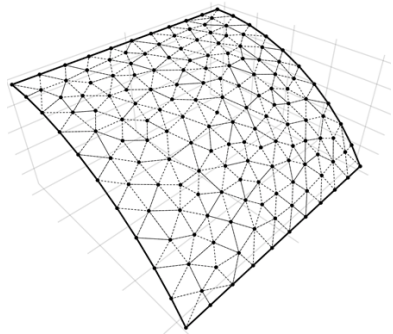
Curved model

Results of patch test for cylindrical model.

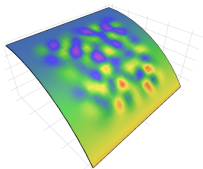
	Linear patch test		Quadratic patch test	
	L_2 -Error	H_e -Error	L_2 -Error	H_e -Error
GI-Penalty	1.75E-04	4.50E-03	1.08E-03	5.83E-03
GI-Nitsche	1.77E-04	5.36E-03	1.07E-03	6.33E-03
RKGSi-Penalty	8.59E-05	9.11E-04	4.28E-04	2.08E-03
RKGSi-Nitsche	1.27E-05	5.32E-05	1.88E-05	5.6E-04
RKGSi-HR	1.43E-05	1.60E-04	2.93E-05	2.85E-04

$$v = \begin{Bmatrix} (\xi^1 + 2\xi^2)^n \\ (3\xi^1 + 4\xi^2)^n \\ (5\xi^1 + 6\xi^2)^n \end{Bmatrix}, \quad n = \begin{cases} 1 & \text{Linear} \\ 2 & \text{Quadratic} \end{cases}$$

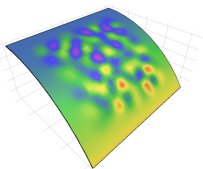
数值算例



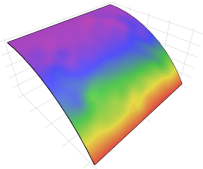
Curved model



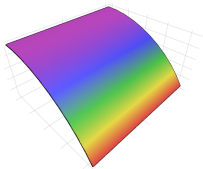
GI-Penalty



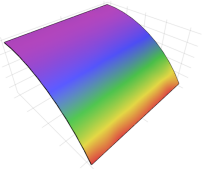
GI-Nitsche



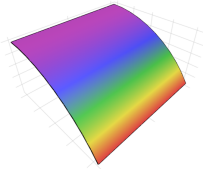
RKGSI-Penalty



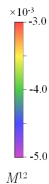
RKGSI-Nitsche



RKGSI-HW

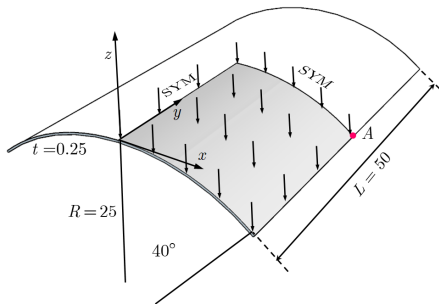


Exact solution



曲壳分片试验 M^{12} 云图

数值算例

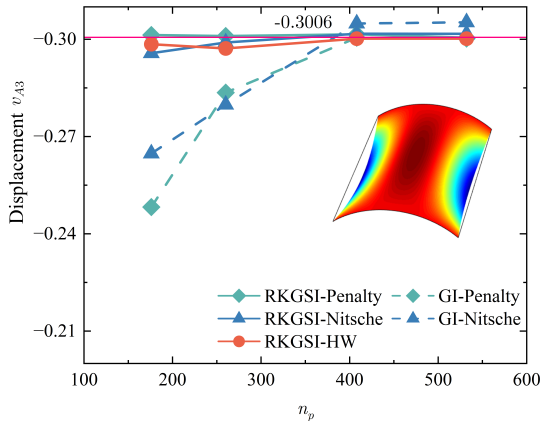


Scordelis-Lo 屋顶问题描述

杨氏模量 $E = 4.32 \times 10^8$

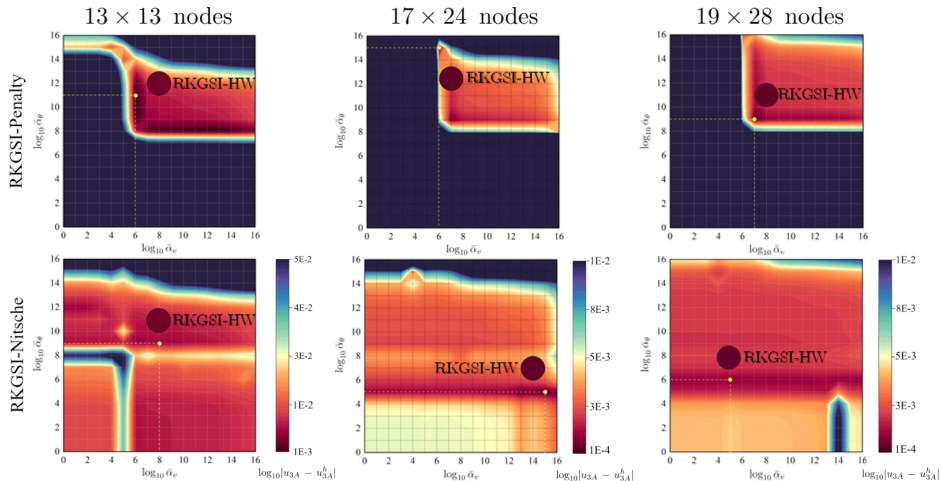
泊松比 $\nu = 0.0$

荷载 $q = 90.0$



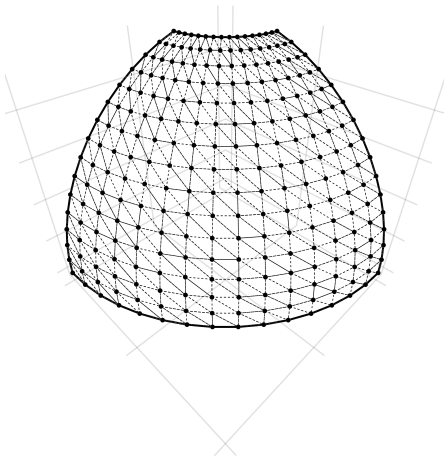
Scordelis-Lo 屋顶问题的位移收敛性

数值算例

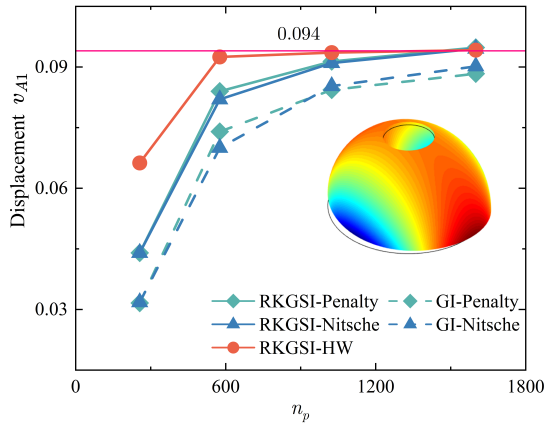


Scordelis-Lo 屋顶问题中参数 α_v 和 α_θ 的灵敏度对比

数值算例

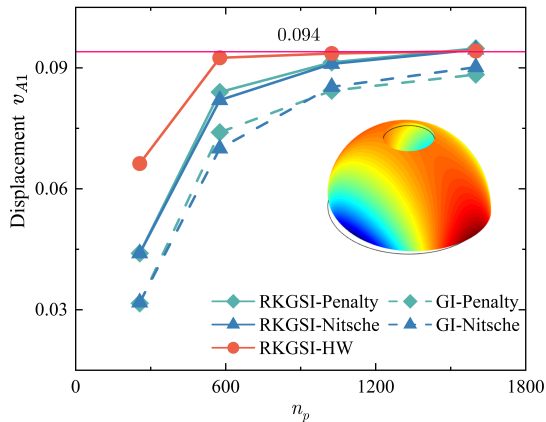
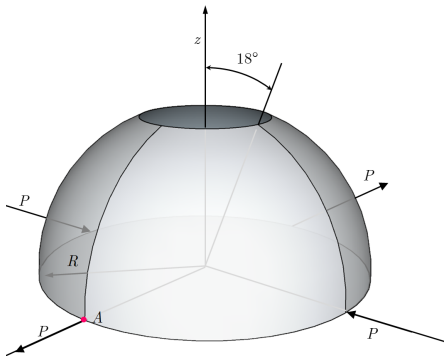


曲壳屋顶问题



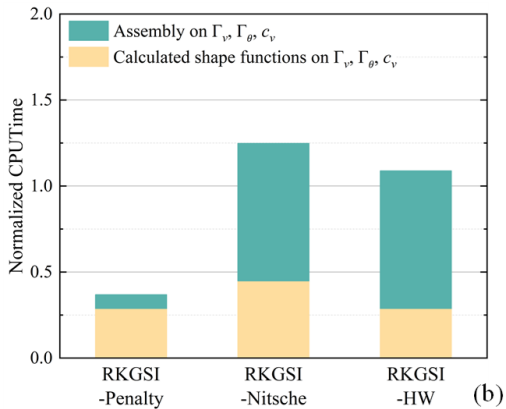
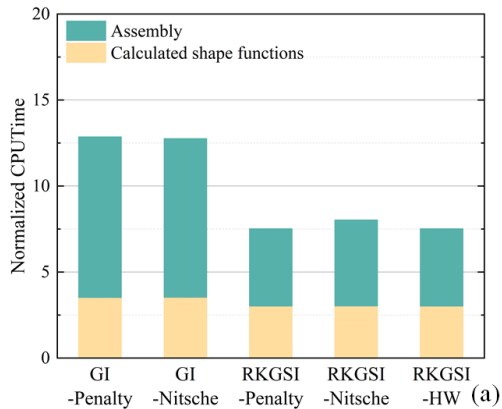
球壳问题

数值算例



球壳问题

数值算例



受压半球壳问题之效率比较：(a) 全域；(b) 本质边界

结论



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